Introduction Of Research Question

The home advantage phenomenon, where teams win more than 50% of games at their home venue, is one of the most robust findings in sports research (Courneya & Carron, 1992). Jamieson's (2010) comprehensive meta-analysis of athletic competitions found that home teams won approximately 60% of contests across various sports, with the effect being particularly pronounced in indoor sports such as basketball and ice hockey (Schwartz & Barsky, 1977). In the National Hockey League (NHL), recent analyses indicate home teams achieve a points percentage of .585 compared to .524 for road teams, representing a 5.5% performance advantage (Sound of Hockey, 2025). However, while extensive research has examined home advantage in overall game outcomes, a critical gap exists in understanding how venue affects specific game situations, particularly special teams play. This literature review synthesizes existing research on home advantage theory, its manifestation in hockey, and special teams performance to establish the foundation for examining powerplay success rates across home and away venues.

Theoretical Frameworks of Home Advantage

Courneya and Carron (1992) established the foundational conceptual framework for understanding home advantage, identifying four primary game location factors that differentiate home and away performance: crowd effects, travel considerations, familiarity with the venue, and rule factors. Building on this work, Carron et al. (2005) proposed a more sophisticated feed-forward model with five interconnected components: game location, game location factors, critical psychological states, critical behavioral states, and performance outcomes. This model posits that game location factors influence both psychological states (such as confidence, anxiety, and arousal) and behavioral states (including skill execution, strategy implementation, and effort expenditure), which ultimately affect performance outcomes. The model's strength lies in its

recognition that home advantage operates through multiple pathways rather than a single mechanism.

The psychological dimension of home advantage has received substantial empirical support. Research on professional soccer players has revealed significant physiological differences between home and away preparation, with athletes exhibiting testosterone levels 40-67% higher before home matches against rivals compared to away games (cited in Sporting Bounce, 2018). These hormonal differences suggest that territorial defense mechanisms activate powerful physiological responses that may enhance performance. From a behavioral perspective, the 'teamwork theory' proposed by Jones (2023) offers an alternative explanation, suggesting that home advantage may be better understood as an 'away disadvantage.' Jones argues that visiting players experience attentional narrowing, an intense focus on opponents due to the perceived threat of the hostile crowd, which impairs their ability to coordinate effectively with teammates. This theory is particularly relevant for hockey, where seamless teamwork and split-second coordination are essential for success, especially during complex special teams situations like powerplay execution.

Home Advantage in Professional Hockey

The magnitude of home advantage in the NHL has varied considerably across eras. Jones (2023) documented that home advantage remained consistently large for approximately 40 years before declining by half between 1991 and 2000, where it has stabilized since. This dramatic shift coincided with two significant rule changes: the introduction of Instant Video Replay (IVR) in 1991-92, which moved goal-mouth decisions from on-site officials to NHL headquarters in Toronto, and the implementation of the instigator rule in 1992-93, which reduced fighting frequency. While Jones initially hypothesized that IVR reduced referee bias favoring home

teams, subsequent analysis of the American Hockey League and East Coast Hockey League, which experienced similar home advantage declines without implementing IVR, contradicted this explanation. The more likely explanation involves the cultural shift toward reduced fighting, which altered the intimidation factor that visiting teams faced. Contemporary analysis by Swartz and Arce demonstrates that home advantage continues to decline over time in both the NHL and NBA, with statistical significance ($p < 0.001$), suggesting ongoing structural changes in how venue affects performance.

Unlike many other sports, hockey provides home teams with explicit rule-based advantages that may amplify venue effects. The 'last change' rule (NHL Rule 82.2) allows the home team's coaching staff to see the visiting team's line configuration during stoppages and adjust their personnel accordingly, enabling favorable matchups. This tactical advantage is particularly significant for special teams, where coaches can deploy their powerplay unit against weaker penalty-killing defenders or match their top offensive threats against opponents' less skilled players. Additionally, faceoff rules (NHL Rule 76.4) require the defending player, often the visiting team in the offensive zone to place their stick on the ice first, providing the home team a marginal timing advantage. Beyond rule factors, Pace and Carron (1992) demonstrated that travel distance correlates with home advantage magnitude, with teams performing worse as travel distance increases. This travel effect likely operates through circadian disruption, accumulated fatigue, and reduced practice time, all of which could impair the precision required for powerplay execution.

Contextual and Situational Moderators

Importantly, home advantage is not uniform across all game situations, and research has identified specific contexts where the advantage diminishes or even reverses. McEwan et al.'s

(2019) analysis of 100 years of NHL playoff overtime games revealed a striking pattern: when away teams had an opportunity to clinch a series, they were significantly more likely to win in overtime compared to home teams facing the same scenario. This 'choking under pressure' phenomenon suggests that supportive crowds can create performance pressure that undermines rather than enhances execution, particularly in high-stakes individual skill situations. Hoffmann et al. (2017) extended this finding by examining home advantage across regulation play, overtime, and shootouts, demonstrating that home advantage patterns vary systematically across game phases. These findings are particularly relevant for powerplay analysis, as special teams situations represent higher-pressure moments where the home crowd's expectations may create additional psychological burden rather than support.

The COVID-19 Natural Experiment

The COVID-19 pandemic provided an unprecedented natural experiment to isolate crowd effects from other home advantage mechanisms. Arboix-Alió et al.'s (2021) analysis of European rink hockey leagues found that home advantage decreased from 63.99% to 57.41% when games were played without spectators, representing a 6.6 percentage point decline. Critically, home advantage did not disappear entirely, suggesting that factors beyond crowd support—such as travel fatigue, venue familiarity, and rule advantages—contribute meaningfully to home performance. This finding has important implications for understanding powerplay differentials, as it suggests that the last change rule and familiarity with ice conditions may independently affect special teams success even when controlling for crowd influence.

Special Teams Performance and the Research Gap

Special teams play occupies approximately 20-25% of game time but exerts disproportionate influence on outcomes (cited in mgoblog). Liu et al.'s (2023) comprehensive

analysis of multiple NHL seasons found that individual powerplay goals carry similar value to even-strength goals, though even-strength play's larger volume makes it the primary contributor to team success over a full season. Importantly, Liu et al. identified a high correlation between teams that perform above or below average in both even-strength and powerplay situations, suggesting that team quality affects both domains similarly. However, their analysis did not examine whether this correlation holds when disaggregating home and away performance. If special teams performance is more sensitive to venue effects than even-strength play, due to tactical advantages like last change or the heightened pressure of man-advantage situations, then home and away powerplay differentials may exceed overall game differentials.

Despite the extensive literature on home advantage and growing research on special teams analytics, a systematic examination of venue effects on powerplay success rates remains absent from peer-reviewed research. Link and Lang's (2024) sophisticated analysis of spatial roles in hockey special teams using NHL tracking data examined power play formations and player positioning but did not investigate whether these tactical structures yield different success rates at home versus away. Hockey analytics blogs such as Hockey-Graphs have explored powerplay zone entry success and formation efficiency, yet these analyses focus on tactical effectiveness without considering venue as a moderating variable. This gap is particularly striking given that the last change rule, an explicit home team advantage, should theoretically provide the greatest tactical benefit during special teams play when matchups are most critical. Therefore, this study addresses a fundamental question: Do NHL teams exhibit significantly different powerplay success rates when playing at home versus away, and if so, which mechanisms (last change advantages, crowd influence, travel effects, or referee bias) best explain the observed differences?

Data & Methodology

Data Sources

Play-by-play data were obtained from the NHL's official reports available on the league's website. The dataset includes detailed information for each power-play opportunity, including date, home or away status, opponent, power-play duration, shot attempts, goals scored, and contextual game situations. Team-level variables, such as regular-season standings, points percentage, and divisional alignment were collected from NHL.com. Travel and scheduling data, including back-to-back games and rest days, were compiled from ESPN and official team schedules to account for fatigue-related factors that may influence performance. Travel distance was calculated using straight-line measurements in miles to represent the shortest possible path between two cities. This approach excludes road networks and detours, providing a standardized baseline for distance without incorporating additional travel to hotels or other locations.

The analysis begins with 2021-22 for several reasons. First, this ensures consistency in league composition, as the NHL has maintained 32 teams since Seattle's expansion. Second, COVID-19 disrupted the 2019-20 and 2020-21 seasons with games without fans, altered schedules, and temporary realignments that would confound home advantage measurements. Third, this modern era features consistent data quality and tracking technology. Finally, focusing on recent seasons ensures findings reflect contemporary NHL rules, strategies, and competitive balance.

Variable Construction

The dependent variable is power-play success, operationalized as power-play percentage (PP%), calculated as the number of power-play goals divided by the total number of opportunities. This metric is assessed at the individual power-play opportunity level, with each

attempt associated with a unique percentage due to the dynamic nature of penalty-kill performance calculated for each instance.

The primary independent variable is venue, coded as a binary indicator (1 = home, 0 = away), capturing factors such as crowd support, last-change rule advantages, travel-related effects, and venue familiarity.

To assess shot quality, an expected goals (xG) model was developed to estimate the probability that a shot results in a goal based on its location. The model incorporates key features, including distance from the goal, shooting angle, and shot location (slot versus perimeter). Expected goals for individual shots were aggregated to compute total xG per power play, providing a measure of both shot quality and shot volume during each opportunity. Model performance indicates good discrimination, with an AUC of 0.71, and well-calibrated predictions, as reflected by a Brier Score of 0.062.

Control variables include schedule factors: back-to-back games (binary indicator), rest days (continuous count of days since last game), and travel distance (measured in miles). Season fixed effects and team fixed effects account for league-wide trends in powerplay success rates and persistent team quality differences.

Analytical Strategy

Analysis proceeds in two stages. First, descriptive statistics compare aggregate powerplay performance between home and away games across all teams and seasons. Visual analysis examines distributions to identify outliers and assess whether differences represent league-wide patterns or are driven by specific teams.

The first regression analysis estimates home venue effects while controlling for confounding factors. The baseline ordinary least squares (OLS) specification is:

$PP\%_i = \beta_0 + \beta_1(\text{Home})_i + \beta_2(\text{xG/Shot})_i + \beta_3(\text{Rest Days})_i + \beta_4(\text{Back-to-Back})_i + \beta_5(\text{Travel Distance})_i + \gamma + \varepsilon_i$

where i indexes powerplay opportunities, β_1 represents the home advantage effect, and γ represents season fixed effects. The coefficient β_1 is the primary parameter of interest, representing the percentage point difference in powerplay success attributable to home venue, holding other factors constant.

Second, temporal trends are examined by comparing home and away powerplay percentages across seasons to test whether declining home advantage in overall outcomes extends to special teams. The baseline ordinary least squares (OLS) specification is:

$PP\%_i = \beta_0 + \beta_1(\text{Home})_i + \beta_2(\text{xG/Shot})_i + \beta_3(\text{Rest Days})_i + \beta_4(\text{Back-to-Back})_i + \beta_5(\text{Travel Distance})_i + \beta_6(\text{Season})_i + \gamma + \varepsilon_i$

where i indexes powerplay opportunities, β_1 represents the home advantage effect, and γ represents season fixed effects. This model introduces season-level separation, allowing analysis to be conducted on a per-season basis rather than aggregating all data into a single combined dataset. The previous model primarily examined overall trends by pooling multiple seasons together, which provided a broad perspective but limited the ability to capture year-to-year variability. By treating each season as a distinct entity, this approach enables a more granular examination of team performance across all 82 games within a season. This distinction is particularly important for assessing the impact of roster changes, such as player acquisitions or departures, as well as coaching adjustments, which can significantly influence power-play effectiveness from one season to the next. Incorporating season as a variable also helps control for league-wide changes, such as rule modifications or shifts in playing style, that may affect power-play success over time.

Hypotheses and Robustness

Based on existing literature, three hypotheses are tested. First, home teams should exhibit higher powerplay success rates, driven primarily by last change rule advantages enabling favorable matchups. Second, shot quality should be the strongest predictor of powerplay success, as higher expected goals per shot indicates better scoring opportunities. Third, fatigue factors such as back-to-back games should negatively affect powerplay performance.

The dataset consists of 40,970 powerplay opportunities across 5,603 games, covering all 32 NHL teams from the 2021-22 through 2024-25 seasons. This substantial sample size provides robust statistical power to detect meaningful venue effects while controlling for confounding variables including home/away status, shot quality metrics, rest days, and travel distance.

Results

Model One

The initial model examined the relationship between power-play success (PP%) and key predictors, including shot quality, venue, rest, scheduling, and travel (Figure 1). Results indicate that shot quality, measured as expected goals per shot (xG/Shot), had the strongest positive effect, with a +5.54 percentage point increase in PP% per unit increase in xG/Shot. Playing at home was also associated with improved performance, contributing an average increase of +1.32 percentage points per home game. Additional rest days provided a minimal benefit (+0.017 per day), while back-to-back games imposed a negative effect (-0.353), suggesting fatigue reduces power-play efficiency. Travel distance exhibited a small positive effect, though its

```
==================================================================
MODEL 1: OLS REGRESSION - PP% ~ Home + xG + Rest Days + Travel
==================================================================
                            OLS Regression Results
==============================================================================
Dep. Variable:                 pp_pct   R-squared:                       0.008
Model:                            OLS   Adj. R-squared:                  0.008
Method:                 Least Squares   F-statistic:                     57.90
Date:                Sun, 16 Nov 2025   Prob (F-statistic):           3.02e-60
Time:                        14:07:17   Log-Likelihood:            -1.4504e+05
No. Observations:               40703   AIC:                         2.901e+05
Df Residuals:                   40697   BIC:                         2.901e+05
Df Model:                           5
Covariance Type:                  HC3
==================================================================================
                     coef    std err          z      P>|z|      [0.025      0.975]
----------------------------------------------------------------------------------
Intercept         25.7045      0.097    263.917      0.000      25.514      25.895
is_home            1.3185      0.087     15.081      0.000       1.147       1.490
xG_per_shot        5.5441      0.913      6.071      0.000       3.754       7.334
rest_days         -0.0167      0.008     -1.991      0.047      -0.033      -0.000
back_to_back      -0.3533      0.143     -2.475      0.013      -0.633      -0.073
travel_distance    0.0002   7.91e-05      3.150      0.002    9.41e-05       0.000
==============================================================================
Omnibus:                     8768.185   Durbin-Watson:                   0.166
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            81472.118
Skew:                           0.774   Prob(JB):                         0.00
Kurtosis:                       9.756   Cond. No.                     1.53e+04
==================================================================
```

Figure 1

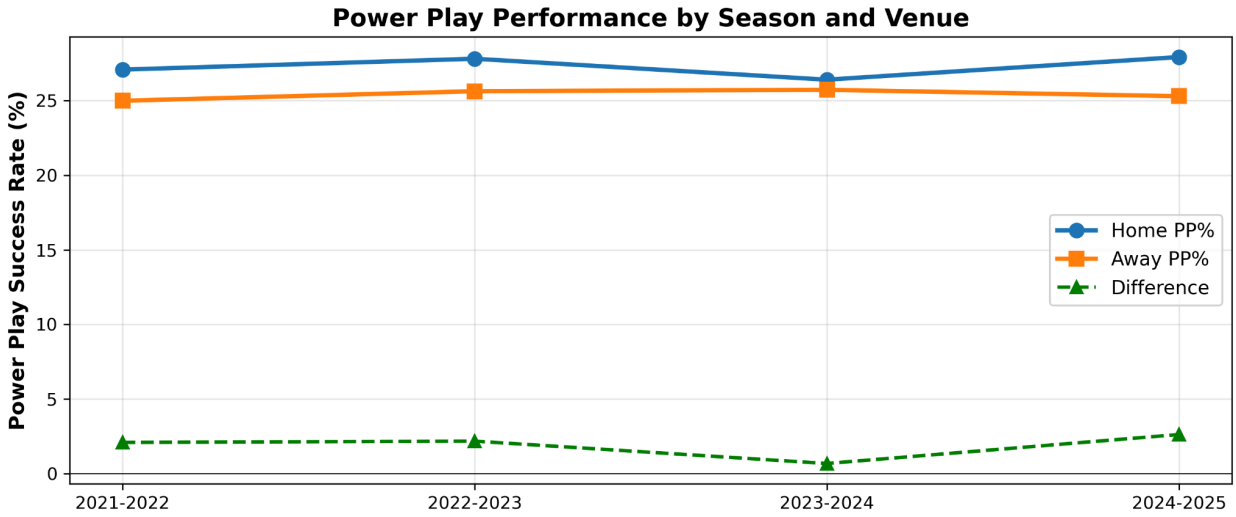
practical significance was limited. Despite these relationships, the model explained very little variance in PP% ($R^2 = 0.008$), indicating that most variation is driven by factors outside the included predictors. However, the large sample size (n = 40,789) supports the reliability of these estimates, even though the impact of rest and travel is negligible at the individual opportunity level.

As shown in Figure 1, the model produced a very low R^2 value (0.008), indicating that the included predictors explain only a small fraction of the variance in power-play success. This outcome is expected given the high variability and inconsistency in individual power-play opportunities, where numerous unobserved factors—such as player skill, puck luck, and real-time tactical decisions—drive results. While the data are noisy at the opportunity level, the large sample size (n = 40,789) ensures that coefficient estimates remain statistically reliable. According to the Central Limit Theorem, as sample size increases, the distribution of the sample mean approaches normality, reducing the impact of random variation and stabilizing parameter estimates. Thus, despite the low explanatory power, the model provides meaningful directional insights into the effects of shot quality, venue, and fatigue-related factors.

Model 2

The second model incorporates season-level controls to account for year-to-year variability in power-play performance. Results indicate that playing at home continues to provide a positive advantage, increasing PP% by approximately 1.31 percentage points. Shot quality (xG/Shot) remains the strongest predictor, with a +7.66 percentage point increase per unit, even after controlling for season effects. Fatigue-related factors persist, as back-to-back games impose a small but negative impact (-0.037), reinforcing the influence of condensed scheduling on special-teams efficiency. Incorporating season effects improved model fit, with adjusted R^2

increasing from 0.008 in the base model to 0.011, demonstrating that season-level variation explains additional variance in power-play success. As shown in Figure 2, home-ice advantage remains consistent across seasons, and the inclusion of season trends provides valuable context for interpreting these effects. These findings highlight the stability of shot quality as a key driver of performance while confirming that venue and fatigue effects remain relevant across multiple seasons.



Season	Home PP%	Away PP%	Advantage
2021-2022	27.09%	24.99%	+2.09%
2022-2023	27.81%	25.64%	+2.18%
2023-2024	26.41%	25.73%	+0.68%
2024-2025	27.93%	25.30%	+2.62%

Figure 2

The second model represents an improvement over the first by introducing season-level controls, which account for year-to-year variability in power-play performance. As shown in Figure 2, season effects explain additional variance in PP% and highlight consistent home-ice advantages across multiple seasons. While the adjusted R^2 remains low (0.011 compared to 0.008 in the base model), this increase demonstrates that incorporating season trends provides a more

nuanced understanding of performance drivers. Importantly, shot quality (xG/Shot) remains the strongest predictor even after controlling for season, reinforcing its central role in power-play success. Venue and fatigue-related effects also persist, suggesting these factors are stable across seasons. Overall, the second model offers greater interpretability and contextual insight, even if predictive power remains modest.

Model Diagnostics

Model diagnostics were conducted to evaluate the assumptions and performance of the regression models. As shown in Figure 3, the residuals versus fitted plot indicates some heteroscedasticity and non-linearity, which is expected given the complexity and variability of in-game hockey data. The Q-Q plot demonstrates that residuals generally follow a normal distribution, with slight deviations at the tails, suggesting minor departures from normality. The actual versus predicted plot confirms the model's limited predictive power, consistent with the low R^2 values reported earlier. Finally, the residual distribution appears approximately normal, supporting the validity of the model despite its modest explanatory strength. Overall, these diagnostics confirm that while the model meets key assumptions reasonably well, its predictive capability is constrained by the inherent randomness and unobserved factors in power-play performance. Including these diagnostics is essential for verifying that the underlying assumptions of ordinary least squares (OLS)—such as linearity, normality of residuals, and homoscedasticity—are reasonably satisfied, ensuring that the reported coefficients and inferences are statistically sound.

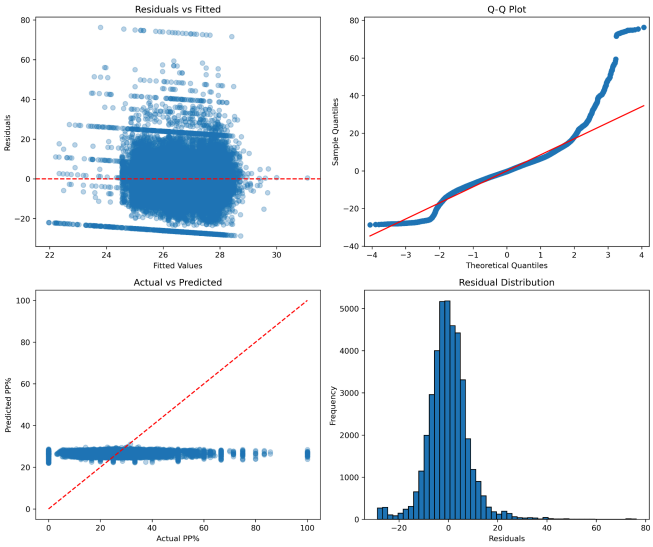


Figure 3

Home vs. Away

As shown in Figure 4, teams perform better on the power play when playing at home, with an average success rate of 27.30% compared to 25.42% on the road. This home-ice advantage of nearly 1.9 percentage points aligns with previous findings in hockey analytics, where factors such as crowd support, last-change rule, and familiarity with rink conditions contribute to improved performance. While the effect size is modest compared to other predictors, it remains consistent across seasons and reinforces the importance of venue in special-teams success.

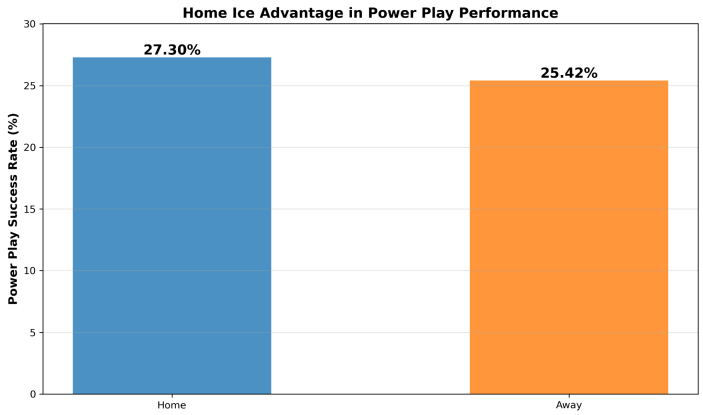


Figure 4

xG Within a Powerplay

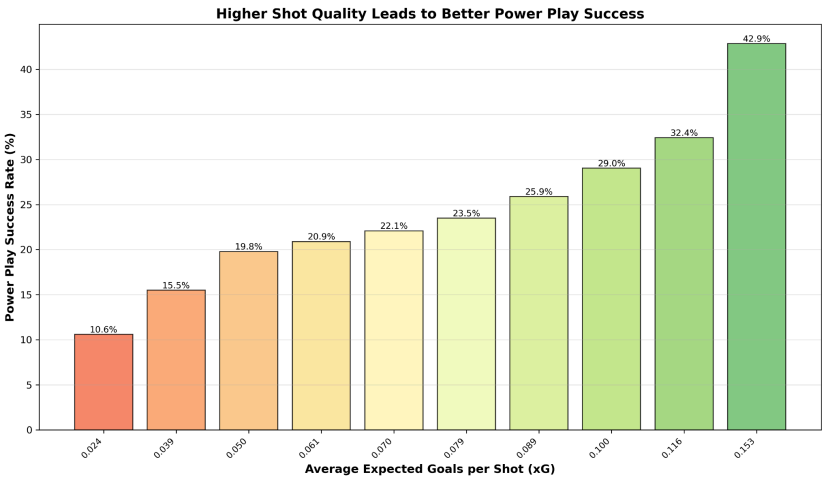


Figure 5

Figure 5 illustrates the strong relationship between shot quality, measured by average expected goals per shot (xG/Shot), and power-play success. Teams generating higher-quality chances see significantly better outcomes, with success rates rising from 10.6% at an average xG of 0.034 to 42.9% at 0.153.

This trend represents an approximate increase of +7.66 percentage points in PP% per unit increase in xG/Shot, making shot quality the most influential predictor in both models. Unlike

venue or fatigue effects, which provide incremental changes, shot quality fundamentally alters scoring probability by improving the likelihood that each attempt becomes a goal.

While quality is critical, volume remains an important complementary factor. A team that consistently generates high-xG shots but in low quantities may still underperform compared to one that combines quality with sustained pressure. Summing expected goals across all shots during a power play provides a holistic measure of opportunity strength, capturing both dimensions. This approach reflects real-world strategy, where elite power plays aim to create multiple high-danger chances rather than relying on a single perfect shot.

The implications for coaching and roster construction are clear: prioritizing players and systems that can create slot-area (high-danger) chances and maintain puck movement under pressure is essential for maximizing power-play efficiency. Teams that fail to adapt and continue generating perimeter shots with low xG values risk falling behind in special-teams performance. These findings underscore the importance of analytics-driven decision-making in optimizing power-play tactics.

Back-to-Back and Travel

As shown in Figure 6 (Panel A), power-play success declines slightly in back-to-back situations, dropping from 26.42% with regular rest to 26.10% when teams play on consecutive nights. While the difference is modest, it reflects the cumulative effect of fatigue on execution and decision-making during special-teams play. Even small reductions in efficiency can be meaningful over the course of a season, particularly for teams that rely heavily on their top power-play unit. This finding reinforces the importance of managing player workload and optimizing line deployment during condensed schedules and limited rest.

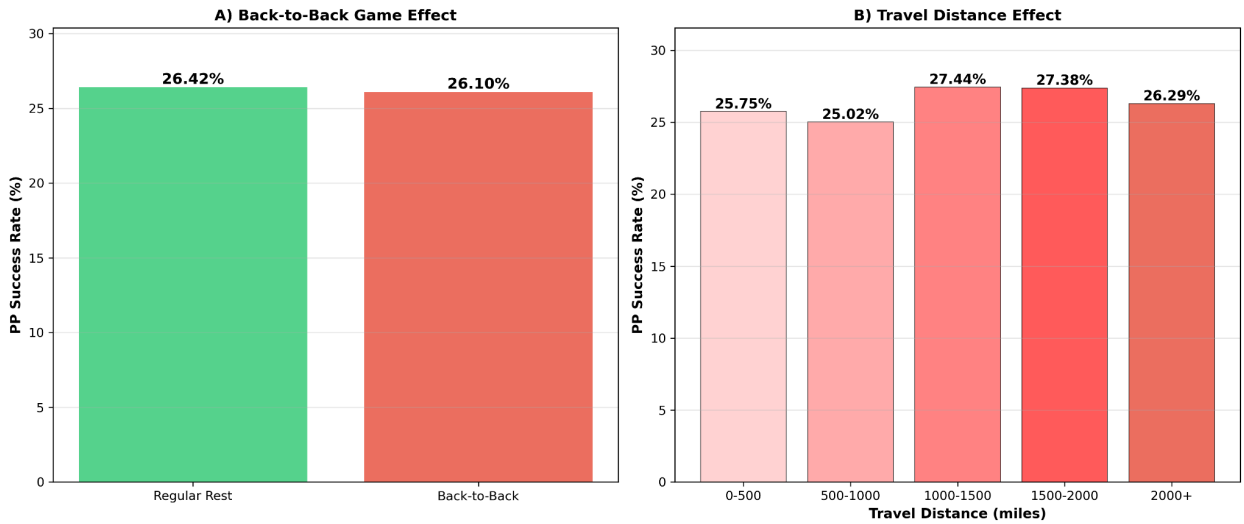


Figure 6

Panel B of Figure 6 illustrates the relationship between travel distance and power-play success, revealing a non-linear pattern. Teams traveling 0–500 miles or 500–1,000 miles exhibit lower success rates (25.75% and 25.02%, respectively), while performance improves significantly for trips in the 1,000–1,500 mile and 1,500–2,000 mile ranges (27.44% and 27.38%). Interestingly, success declines again for distances beyond 2,000 miles (26.29%), suggesting that extreme travel introduces fatigue and logistical challenges that offset potential benefits. These results indicate that moderate travel distances may actually correlate with better power-play outcomes compared to short or very long trips.

The dip in performance at shorter travel ranges likely reflects increased opponent familiarity rather than physical fatigue. Teams within 1,000 miles of each other generally play three to four times per season, creating a larger sample of shared game film and tactical exposure. This familiarity allows penalty-killing units to anticipate power-play setups and neutralize preferred shooting lanes more effectively. In contrast, teams facing opponents from greater distances encounter less frequent matchups, reducing the ability to prepare for specific tendencies. Consequently, while travel introduces logistical challenges, unfamiliarity with distant

opponents may provide a strategic advantage that offsets fatigue, explaining the observed improvement in power-play success at intermediate travel ranges.

Largest Differences

As shown in Figure 7, the top five team-seasons with the largest home versus away power-play percentage differences highlight how venue effects can vary dramatically across teams. The Columbus Blue Jackets (2024–2025) lead the list with an extraordinary 18.21 percentage point gap, followed by the Colorado Avalanche (2024–2025) at 13.68 points and the Ottawa Senators (2024–2025) at 13.31 points. The Buffalo Sabres (2021–2022) and Ottawa Senators (2022–2023) round out the top five with differences of 11.85 and 11.26 points, respectively. These gaps are far above the league-wide average of roughly 1–2 percentage points (Figure 4), indicating that certain teams derive substantial benefits from playing at home. Such disparities suggest that home-ice advantage is not uniform across the league but can be amplified by team-specific factors such as roster composition, tactical systems, and coaching strategies.

Both the Ottawa Senators (2022–2023) and the Buffalo Sabres (2021–2022) had an eleven-percentage-point difference between their home and away power-play success rates, highlighting the inconsistency often found in rebuilding teams. During these seasons, Tim Stützle and Tage Thompson emerged as breakout stars, each scoring more than 25% of their total goals on the power play and becoming central pillars of their teams' offensive resurgence. In contrast, the Colorado Avalanche in 2024–2025 remained a top-tier offensive team, but in Game 7 against the Dallas Stars their power play struggled on the road. Colorado failed to capitalize on any of its three power-play opportunities, a missed chance that contributed to losing a 2–0 lead and ultimately falling 4–2. Even one conversion might have shifted the game's momentum and altered the trajectory of their playoff run. Overall, the contrast between rebuilding teams like

Ottawa and Buffalo and established contenders like Colorado underscores a central truth in modern NHL competition: special-teams performance often determines postseason survival. While young, developing teams may lean heavily on emerging stars and benefit from home-ice advantages to mask structural inconsistencies, elite contenders cannot afford similar volatility. For teams with championship aspirations, maintaining a reliable, portable power play—one that performs consistently regardless of venue—is essential. Colorado's Game 7 collapse illustrates how quickly momentum can swing when special teams fail, and it emphasizes that in tight playoff series, a single missed power-play opportunity can be the difference between advancement and elimination.

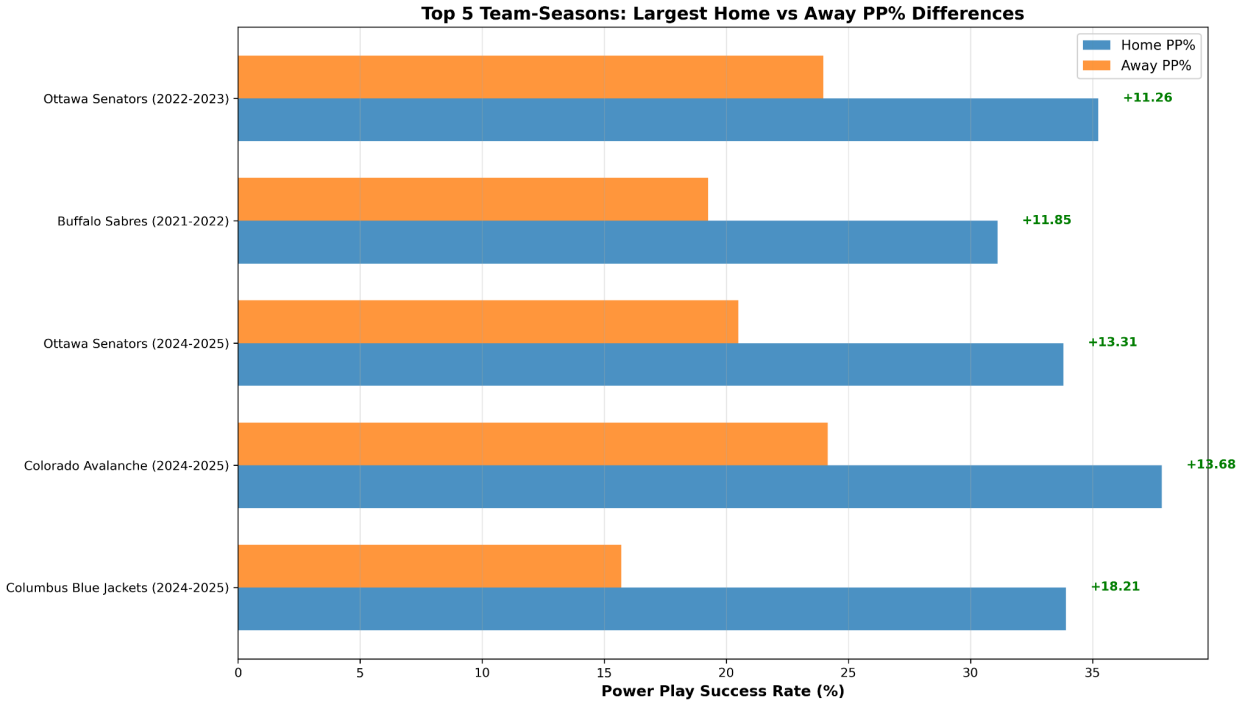


Figure 7

The power play remains an integral component of NHL success, and the difference between home and away performance can meaningfully shape game outcomes. A balanced power-play unit reduces variance across venues, minimizes momentum loss in hostile

environments, and increases a team's probability of sustained success. As this analysis demonstrates, achieving consistency in special-teams execution is not merely advantageous—it is a competitive necessity for teams seeking deeper postseason runs and long-term progression.

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